UNIT 2

Theoretical Structure of Quantum Information Systems

Unit 2 at a glance

Qubit (spin & polarization) · bits vs qubits · physical systems (trapped ions, superconducting, photons) · coherence & decoherence · Hilbert space, states, operators (abstract) · entanglement & non-locality · quantum vs classical information · philosophical implications

1. What is a Qubit?

Classical information is stored in a bit, which is always either 0 or 1. Quantum information is stored in a qubit (quantum bit): any two-level quantum system whose two basis states are labelled $|0\rangle$ and $|1\rangle$. Unlike a bit, a qubit can be in a superposition of both:

Qubit state

$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$

α, β = complex probability amplitudes. $P(0) = |\alpha|^2$, $P(1) = |\beta|^2$.

Equal superposition: $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ → 50 % chance of each result.

When we measure a qubit we never see 'a bit of both' - we get a single classical bit, 0 or 1, with the above probabilities. Before measurement the amplitudes behave like waves and can interfere. The art of quantum computing is to make wrong answers cancel and right answers reinforce.

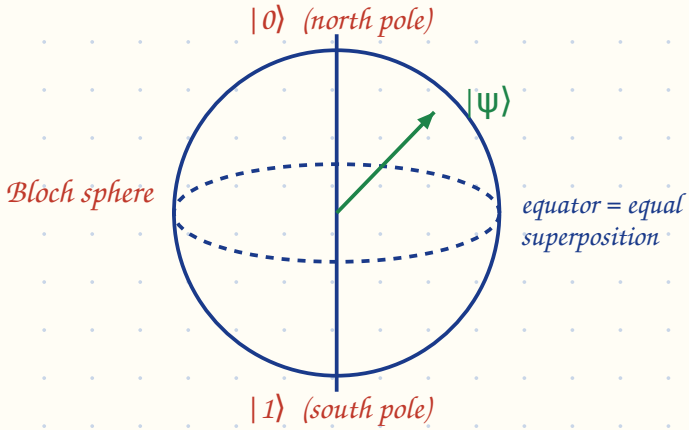


Fig 1: Bloch sphere - geometric picture of one qubit

» Bloch sphere picture

- Every pure single-qubit state is a point on the surface of a sphere of radius 1.
- North pole = $|0\rangle$, south pole = $|1\rangle$; points on the equator are equal superpositions with different relative phase.
- $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle$ - θ is the latitude angle, φ the longitude (phase).

- Quantum gates = rotations of the arrow on this sphere. Noise = arrow shrinking/drifting.

2. Physical Realizations: Spin and Polarization



Fig 2: Two natural two-level systems

» 2.1 Spin qubit (electron or nucleus)

- Spin is an intrinsic angular momentum, like a tiny magnet. For spin-½, measurement along an axis gives only two results: up (↑) or down (↓).
- Map: $|\uparrow\rangle \to |0\rangle$, $|\downarrow\rangle \to |1\rangle$. A magnetic field splits the two energy levels; microwave/RF pulses rotate the spin.
- Measuring along z gives ↑/↓; measuring along x gives left/right - 'incompatible' directions: an up state is a 50-50 superposition of left & right (uncertainty principle at work).
- Realizations: electrons in quantum dots, phosphorus in silicon, NV centres in diamond, nuclear spins (NMR).

» 2.2 Polarization qubit (photon)

- Polarization = direction of oscillation of light's electric field. Choose H (horizontal) = $|0\rangle$, V (vertical) = $|1\rangle$.
- Diagonal $D = (|H\rangle + |V\rangle)/\sqrt{2}$, anti-diagonal $A = (|H\rangle - |V\rangle)/\sqrt{2}$; circular R/L also possible.
- Manipulated with wave plates ($\lambda/2$, $\lambda/4$), polarizing beam splitters; measured by polarizers + single-photon detectors.
- Test: a polarizer at angle θ passes a photon with probability $\cos^2\theta$ (Malus's law at single-photon level).

> **Exam tip / Remember**
> Spin and polarization are mathematically the same: a two-dimensional state space. Learn this analogy - examiners love it: spin up/down ↔ polarization H/V ↔ $|0\rangle/|1\rangle$.

3. Classical Bits vs Quantum Bits



Fig 3: A bit has 2 states; a qubit has infinitely many

Property	Classical bit	Qubit
Possible states	0 or 1	$\alpha \|0\rangle + \beta \|1\rangle$ (continuum)
Physical carrier	Voltage, magnetic domain, transistor	Spin, photon, ion, superconducting circuit
Reading it	Does not disturb the value	Measurement collapses state; gives 0/1 randomly
Copying	Easy (fan-out)	Impossible for unknown states (no-cloning)
n units describe	1 value out of 2^n at a time	2^n amplitudes at once (superposition)
Correlation	Classical (shared randomness)	Entanglement - stronger than classical
Error behaviour	Digital: robust, simple error correction	Analog-like errors; needs quantum error correction
Operations	AND, OR, NOT (irreversible)	Unitary gates (reversible) + measurement
Stability	Stable for years	Fragile: microseconds to seconds

» Growth of information: why n qubits are special

n classical bits hold ONE of 2^n strings. n qubits can be in a superposition of ALL 2^n strings with an amplitude for each. 10 qubits → 1024 amplitudes; 50 qubits → about 10^{15}; 300 qubits → more numbers than atoms in the universe. A classical computer would need exponentially large memory to store such a state. But beware: measurement returns only n bits, so extracting useful results needs clever algorithms.

★ *Common myth*

A quantum computer does NOT simply try all answers at once and read the best one. Interference must be designed so the right answer is likely at measurement.

4. Physical Quantum Systems for Qubits (Non-engineering View)

Platform	What is the qubit?	How controlled	Good / Bad
Superconducting circuits	Two lowest energy states of an electrical circuit (transmon) cooled to ~15 mK; behaves like an 'artificial atom'	Microwave pulses; resonators for read-out	Fast gates (ns), chip-based, scalable / short coherence (~100 μs), needs dilution refrigerator
Trapped ions	Two internal energy levels of a single charged atom (e.g. Yb^+, Ca^+) held in vacuum by electric fields	Laser pulses; ions share motion to interact	Highest fidelity, long coherence (seconds-minutes), all-to-all links / slow gates, scaling to large ion chains is hard
Photons	Polarization, path or time-bin of a single particle of light	Beam splitters, phase shifters, detectors	Room temperature, travels far, ideal for communication / photons don't interact easily, losses

Other platforms: neutral atoms in optical tweezers, spins in silicon/quantum dots, NV centres in diamond, topological (Majorana, still experimental). All satisfy the same abstract qubit rules - the platform differs only in how well it can be controlled.

5. Quantum Coherence and Decoherence (Intuitive)

Coherence means the superposition keeps a well-defined phase relationship between its parts - like two perfectly synchronized waves - so interference is possible. Decoherence is the loss of that phase relationship because the qubit unintentionally interacts (exchanges energy/information) with its environment: stray photons, heat, magnetic noise, vibrations, neighbouring qubits.

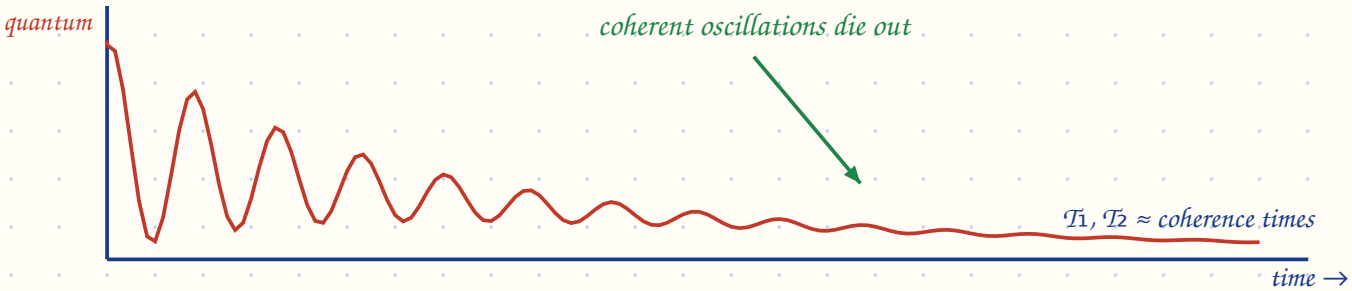


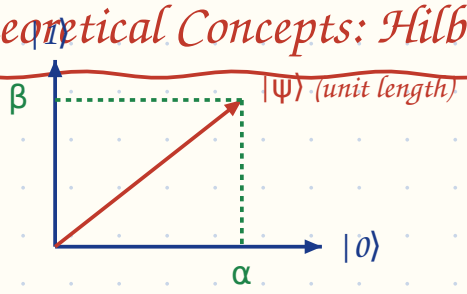
Fig 4: Quantum behaviour fades with time as the environment 'listens'

- Intuition: imagine a spinning coin balanced on its edge. Any slight bump makes it fall - the environment 'measures' it. The more particles it touches, the faster it falls.
- T_1 (relaxation time): time for $|1\rangle$ to decay into $|0\rangle$ by losing energy.
- T_2 (dephasing time): time for the phase between $|0\rangle$ and $|1\rangle$ to be randomized; always $T_2 \leq 2T_1$.

- *Gate operations must finish ≪ T_2. The figure of merit is (coherence time) ÷ (gate time) = number of operations possible.*
- *Bigger systems decohere faster - explaining why the everyday world looks classical (Zurek's decoherence theory).*

> *Fighting decoherence*
> *Isolation (vacuum, shielding, cryogenics) · better materials · dynamical decoupling (echo pulses) · decoherence-free subspaces · quantum error correction.*

6. Theoretical Concepts: Hilbert Space, States, Operators (Abstract)



state = vector
operator = matrix/rotation

measurement = project
then read length²

Fig 5: A state is a unit vector in an abstract space

» 6.1 Hilbert space

A Hilbert space is the abstract mathematical 'stage' for a quantum system: a complex vector space with an inner product (a way to measure lengths and angles/overlaps). Each physical system has its own Hilbert space; a qubit lives in a 2-dimensional one ($\mathbb{C}^2$). Think of it as the set of all allowed 'directions' the system's state can point.

» 6.2 Quantum states

- *A pure state = a unit-length vector $|\psi\rangle$ in the space (rays differing by an overall phase are physically identical).*
- *The inner product $\langle\varphi|\psi\rangle$ is the 'overlap amplitude'; $|\langle\varphi|\psi\rangle|^2$ = probability of finding φ if the system is in ψ.*
- *Basis states: a set of mutually perpendicular unit vectors ($|0\rangle$, $|1\rangle$). Any state = combination of them.*
- *A mixed state (density matrix ρ) = statistical mixture of pure states - describes ignorance or entanglement with something else.*

» 6.3 Operators

- *Operators act on states: $\hat{A}|\psi\rangle = |\psi'\rangle$. In practice they are matrices.*

- *Unitary operators* ($U^\dagger U = I$) = reversible rotations preserving length: all quantum gates and time evolution.
- *Hermitian operators* = observables (energy, spin, position): their eigenvalues are possible measurement results; eigenvectors are the states with those definite values.
- Non-commuting operators ($AB \neq BA$) cannot be measured sharply together → uncertainty principle.

Key operators (just remember the roles)
X = bit-flip: $|0\rangle \leftrightarrow |1\rangle$ · Z = phase-flip: $|1\rangle \to -|1\rangle$ · H (Hadamard) = makes superposition: $|0\rangle \to (|0\rangle + |1\rangle)/\sqrt{2}$.
Tensor product $\otimes$ combines systems: 2 qubits live in a 4-dimensional space $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.

7. The Role of Entanglement and Non-locality

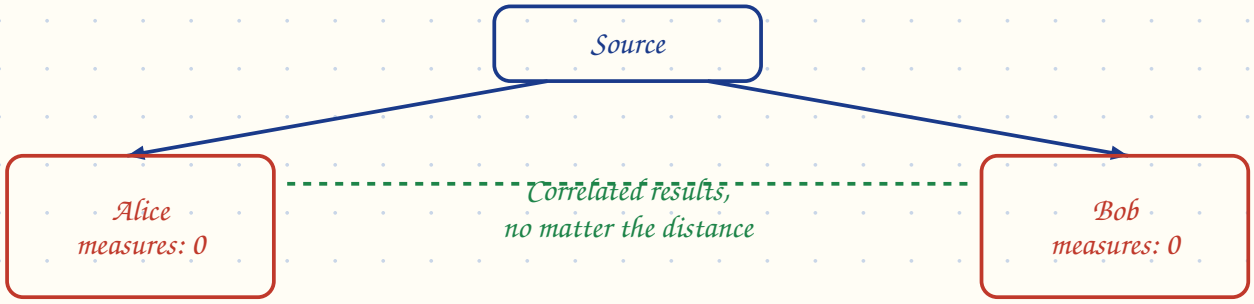


Fig 6: Entangled pair - perfectly correlated outcomes

Two qubits are *entangled* if their joint state cannot be written as a product of separate states. The classic example is the Bell state:

Bell state
$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$
Measure qubit 1: get 0 or 1 with 50 % each. Then qubit 2 is always found with the SAME result. Yet neither qubit alone has a definite value before measurement.

- *Non-locality*: Bell's inequality tests show correlations stronger than any local-hidden-variable theory can give (CHSH bound 2 vs quantum $2\sqrt{2} \approx 2.83$). Loophole-free experiments (2015) confirmed this.
- No faster-than-light signalling - Alice's outcomes alone look random; the correlation shows up only when results are compared via ordinary communication.
- Role in quantum information: *teleportation* (transfer a state using entanglement + 2 classical bits), *superdense coding* (send 2 classical bits with 1 qubit + entanglement), QKD (E91), and the speed-up of quantum algorithms (entanglement lets qubits' states depend on each other).

Monogamy of entanglement

If two qubits are maximally entangled, neither can be entangled with a third. This is why eavesdroppers are detectable in entanglement-based cryptography.

8. Quantum Information vs Classical Information

Aspect	Classical information	Quantum information
Unit	bit (0/1)	qubit; also qudit, continuous variables
Copying	Freely copied, broadcast	No-cloning theorem forbids copying unknown states
Reading	Non-destructive, repeated	Measurement collapses; one copy gives ≤ 1 bit (Holevo bound)
Deleting / hiding	Can be deleted irreversibly	No-deleting theorem; evolution is reversible
Correlations	Local; shared randomness	Entanglement, non-local
Teleporting	Meaningless - just copy	Teleportation: moves the state; original destroyed
Security	Computational assumptions (RSA)	Laws of physics (QKD)
Theory	Shannon entropy $H = -\Sigma p \log p$	von Neumann entropy $S = -Tr\ \rho \log \rho$
Errors	Bit flips only	Bit flips, phase flips, continuous drifts

Three golden theorems

No-cloning: unknown quantum state cannot be copied. No-deleting: unknown state cannot be erased given a copy. No-broadcasting: cannot send the state to many users while keeping it intact. Together they make quantum information special and secure.

9. Philosophical Implications

» Randomness vs determinism

Classical physics (Laplace's demon) is deterministic: know the present exactly and you can compute the future. In quantum mechanics the state evolves deterministically (Schrödinger equation) but individual measurement outcomes are fundamentally random - only probabilities are lawful. Quantum random number generators exploit this to produce genuine, not pseudo-, randomness.

» The role of the observer

- Measurement links the quantum and the classical - the 'measurement problem': why does the state collapse to ONE outcome?
- Copenhagen view: wavefunction = knowledge; collapse upon measurement; the cut between system and apparatus is pragmatic.
- Many-Worlds (Everett): no collapse; all outcomes occur in branching universes.
- Pilot-wave (de Broglie-Bohm): deterministic hidden variables, but non-local.
- QBism / information-based views: quantum states are an agent's degrees of belief.
- Decoherence explains why interference vanishes but not, by itself, why we see one outcome.

» Realism and locality

Einstein-Podolsky-Rosen (1935) asked if quantum theory is complete. Bell (1964) showed that no theory that is both local and realistic can match quantum predictions. So we must give up locality, realism, or both - a deep philosophical shift. Yet the rules are consistent and extremely well tested.

> **Exam tip / Remember**
> For essay questions: present (i) determinism vs randomness, (ii) the measurement/observer problem with 2-3 interpretations, (iii) non-locality (EPR/Bell), (iv) practical payoff - randomness and security used in technology.

9A. Measurement Bases and Basis Dependence

A qubit can be measured in different bases. The state $|0\rangle$ is certain in the Z-basis $\{|0\rangle, |1\rangle\}$ but is a coin-toss in the X-basis $\{|+\rangle, |-\rangle\}$. So information about 'which basis' matters - the foundation of BB84 cryptography.

Basis	States	Spin picture	Photon picture
Z (computational)	$\|0\rangle$, $\|1\rangle$	up / down	H / V (0° / 90°)
X (Hadamard)	$\|+\rangle$, $\|-\rangle$	right / left	D / A (+45° / −45°)
Y (circular)	$\|R\rangle$, $\|L\rangle$	in / out	right / left circular

- Measuring in the wrong basis gives random results and disturbs the state.
- Complementary (mutually unbiased) bases: a state definite in one is maximally uncertain in the other.
- Rule of thumb: the more sure you are about the Z-value, the less you know about X-value - uncertainty principle again.

9B. Pure vs Mixed States (Density Matrix - Idea Only)

Pure state	Mixed state
Complete knowledge: a single vector $\|\psi\rangle$	Partial knowledge: a probabilistic mixture of states
Point on surface of Bloch sphere	Point inside the sphere (centre = completely random)
Superposition: coherent, shows interference	Mixture: classical ignorance, no interference
Example: $\|+\rangle = (\|0\rangle + \|1\rangle)/\sqrt{2}$	Example: 50 % $\|0\rangle$ and 50 % $\|1\rangle$ (a noisy coin)

Important subtlety: a half of an entangled pair, looked at alone, is a maximally mixed state even though the full pair is in a pure state. Decoherence is exactly this process - the qubit becomes entangled with the environment, so on its own it looks like a mixture.

9C. Two Protocols that Show Quantum Information at Work

» Quantum teleportation (steps)

1. Alice and Bob share an entangled pair (one qubit each).
2. Alice has an unknown qubit $|\psi\rangle$ to send. She performs a joint (Bell) measurement on $|\psi\rangle$ and her half of the pair.
3. She gets 2 classical bits (00, 01, 10 or 11) and tells Bob by ordinary communication.
4. Depending on the bits, Bob applies I, X, Z or XZ to his qubit - it now equals $|\psi\rangle$.
5. Original $|\psi\rangle$ is destroyed (consistent with no-cloning). No matter has moved, no faster-than-light signal.

» Superdense coding

With a shared entangled pair, Alice can send two classical bits by transmitting only one qubit: she applies I, X, Z or XZ to her half and sends it; Bob does a Bell measurement on both halves and reads 2 bits. Teleportation and superdense coding are 'mirror images' of each other.

> Resources table
>
> Teleportation: 1 ebit + 2 cbits → 1 qubit. Superdense coding: 1 ebit + 1 qubit → 2 cbits. (ebit = shared entangled pair, cbit = classical bit)

9D. Formula & Concept Sheet

Concept	Formula / statement
Qubit	$\|\psi\rangle = \alpha\|0\rangle + \beta\|1\rangle$, $\|\alpha\|^2 + \|\beta\|^2 = 1$
Bloch form	$\cos(\theta/2)\|0\rangle + e^{i\varphi}\sin(\theta/2)\|1\rangle$
Born rule	$P(i) = \|\langle i\|\psi\rangle\|^2$
Hadamard	$H\|0\rangle = \|+\rangle$, $H\|1\rangle = \|-\rangle$
Two-qubit space	dimension $2^2 = 4$; n qubits $\rightarrow 2^n$
Bell state	$(\|00\rangle + \|11\rangle)/\sqrt{2}$
Unitary	$U^\dagger U = I$ (reversible)
CHSH	Classical ≤ 2; Quantum $\leq 2\sqrt{2}$
Coherence	$T_2 \leq 2T_1$
No-cloning	No U with $U\|\psi\rangle\|0\rangle = \|\psi\rangle\|\psi\rangle$ for all ψ

10. Solved Examples

Example 1

Qubit $|\psi\rangle = (1/\sqrt{5})|0\rangle + (2/\sqrt{5})|1\rangle$. $P(0) = 1/5 = 20\ \%$, $P(1) = 4/5 = 80\ \%$. Normalized since $1/5 + 4/5 = 1$.

Example 2

How many complex numbers describe a general 3-qubit state? $2^3 = 8$ amplitudes (minus 1 for normalization and 1 for global phase $\Rightarrow$ 14 real parameters).

Example 3

Apply Hadamard to $|1\rangle$: $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2} = |-\rangle$. Measurement probabilities still 50/50, but the sign (phase) differs - detectable by a second H, which returns $|1\rangle$ with certainty. Shows interference!

Example 4

Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$: $P(00) = ½$, $P(11) = ½$, $P(01) = P(10) = 0$ - outcomes always agree.

11. Revision Questions

1. Define a qubit. Explain with spin and polarization.

2. Compare classical bits and qubits in a table.
3. Describe trapped-ion, superconducting and photonic qubits.
4. What are coherence and decoherence? Define T_1 and T_2.
5. Explain Hilbert space, states and operators in abstract terms.
6. What is entanglement? Explain non-locality and its role in quantum information.
7. State the no-cloning theorem and its importance.
8. Discuss philosophical implications: randomness, determinism, observer.

12. Short Answer Practice (Model Answers)

Q1. Why can't we clone a qubit?

Quantum evolution is linear (unitary). A cloning machine that copies $|0\rangle$ and $|1\rangle$ would, by linearity, map $(|0\rangle + |1\rangle)/\sqrt{2}$ to an entangled state, not to two copies. Hence no device can copy an arbitrary unknown state. Consequence: eavesdroppers cannot copy photons in QKD.

Q2. Why is a qubit more than a probabilistic bit?

A probabilistic classical bit is a mixture with real probabilities p, 1−p. A qubit has complex amplitudes with phases, so contributions can cancel (destructive interference) - something probabilities never do. Also several qubits can be entangled.

Q3. Why must quantum gates be unitary?

Unitarity preserves total probability (norm = 1) and makes operations reversible. Any non-unitary change (loss, noise) is treated as an error.

Q4. Does entanglement allow faster-than-light communication?

No. Each party's outcomes are random; correlations are visible only after comparing results over a classical channel limited by light speed.

Q5. Which qubit platform is best?

No single winner. Superconducting: fast and chip-friendly. Trapped ions: highest accuracy and connectivity. Photons: best for communication and room temperature. Choice depends on the task and the trade-off between speed, coherence and scalability.

Q6. What is the measurement problem?

Quantum theory has two rules - smooth unitary evolution and abrupt collapse on measurement. Why/when collapse occurs and what counts as a measurement is unresolved; interpretations (Copenhagen, Many-Worlds, Bohm, QBism) offer different answers while predicting the same numbers.

★ *Final summary*

Qubit = two-level system with complex amplitudes → lives in Hilbert space → evolves by unitary operators → fragile to decoherence → entanglement links qubits non-locally → quantum information obeys no-cloning, reversibility and collapse rules → deep questions of randomness and observer.